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Information spreading in a population modeled by continuous asynchronous probabilistic cellular automata

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ABSTRACT

In this paper, we propose a model for information propagation in a population based on cellular automata. Different to what is commonly used in the models of the area, instead of a binary level for the information, individuals have a level of knowledge. Moreover, the population can have a marketing campaign to help to spread the information with a certain limit due to the consideration of marketing rejection in the model. Numerical simulations show that these campaigns must be wisely set in order to not saturate the population and decrease the marketing balance due to the rejection. The simulation data is statistically analyzed by using principal component analysis in order to identify the most relevant variables for the model output. The conclusion is that dealing with this rejection is difficult, as well as choosing the percentage of the population which will receive the marketing, and along with the weight for the word-of-mouth were the most important variables of the model according to the simulations and the principal component analysis.

1. Introduction

Rumor spreading has reached another level of magnitude in the past few years due to internet access, social networks and cell phones available to more people [1]. Had rumor spreading based on regular models for population and SIR model for information been enough, it is evident that we now have a more complex and stochastic situation, which requires proper models to help to identify the system properties.

These models help to describe and simulate the processes in order to control an undesirable rumor to spread, or in other cases, boost an advertisement through social networks. For instance, during the nuclear incident in Japan in 2011, a rumor spread across China that the iodine of the salt would help to prevent health problems related to the nuclear leakage. It caused social panic in China, with salt price raising to ten times its regular price [2]. During a crisis like this, false information can make the situation worst in up to 65% of the cases, according to the report of the National Advisory Commission on Civil Disorder, from US [3]. Four are the main components for false information to spread quickly: the information uncertainty, its importance, the anxiety of the individual knowledge about the fact and last but not least, the willingness to believe the rumor [4].

Therefore, from first models and statistics analysis [3], researchers had to deal with robust mathematical tools to fully understand the dynamics of information spreading. It started relying on the epidemiological SIR model of Kermack–McKendrick [5], where individuals are in one of the following states: *Susceptible*, when individuals are subjected to infection, *Infected* are the ones who have the disease and can transmit to the susceptible, and *Removed*, which are the individuals who got cured of disease. For the information process, susceptible is the individual that does not know about a rumor and can be informed by an infected, and the removed knows about the rumor but does not spread it. This kind of initial approach has been used for modeling a consumer market [6], to model the rumor spreading on Twitter [7–9], general social networks [2,10,11] and to evaluate the main nodes for the spreading on social networks [12,13]. Moreover, variations of SIR model, like SI [7], SIS [14], SCIR [7] and SIHR [2] are also used. Other mathematical tools used for modeling information spreading are ordinary differential equations [1,2], complex networks [2,10,12,15–18], multi-agent systems [7,19], and cellular automata [20–22].

However, most of the works presented assume a binary situation where an individual knows the information or not. Here, we propose a model where the individual has a level of information I, continuous in the interval $0 \leq I \leq 1$ and the processes that change the information level are: learning through word-of-mouth [2,3,18], marketing campaigns [23,24], considering forgetfulness [3,7,18] and negative effects of marketing [25]. Also, we propose a marketing model based on how individuals differently absorb information in campaigns [24], how campaigns are set [26], and what happens when marketing is saturated, turning into rejection for the information.

Marketing campaigns are designed by considering many external factors to the campaign. The word-of-mouth process may play a significant role in boosting the campaigns [27], increasing loyalty when the

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information is positive, and increasing disloyalty when the information being spread is negative about a company or product [28–30]. Besides word-of-mouth, space, time, and content broadcasted must be precisely delineated in order to avoid intrusive campaigns and marketing clutter [31,32]. Therefore, it is essential to make the right choices and understand the possible impacts of a marketing campaign.

All the parameters of the model are analyzed with numerical simulation in order to verify the most relevant for the processes of gaining and losing information, also, to provide a formal statistical view of the simulation results, we use a modified Principal Component Analysis (PCA). The conclusion is that rejection, the percentage of the population which will receive the marketing, and the weight for the word-of-mouth were the most important variables of the model according to the simulations and the principal component analysis.

This paper is organized as follows: in the next section, the mathematical modeling basics and statistical analysis to be used are presented. The model is exhibited in Section 3 with the simulation results on Section 4. A final discussion is on Section 5.

2. Mathematical modeling and statistical analysis

2.1. Cellular automata

The Cellular Automata (CA) has been proposed by John von Neumann and Stanislaw Marcin Ulam to model self-reproducing organisms and cellular growth, in the 1940s. A CA is a computational mathematical tool to represent physical systems composed by discrete parts with local interactions. Such interactions may change the cells states, and CA evolves. The classical properties to fully define a CA are [33]:

1. *Shape*: cells structure, with one, two or three dimensions, for instance. The elementary cellular automata studied by S. Wolfram [33] has one dimension, populations usually are modeled by a two-dimensional lattice with periodic boundary condition [34];
2. *Cells states*: CA cells have a finite number of states. The elementary CA has two states, and the classical SIR model, three;
3. *Transition rules*: rules for cell states time evolution, that is, the cell state at $t + 1$ time step is a function of some characteristics of the time step t;
4. *Neighborhood*: cells set connected to each cell of the lattice. In the case of the elementary CA, cells have two neighbors. A Moore neighborhood for the two-dimensional lattice has eight neighbors.

Other characteristics which also define an CA and were used in more recent works are:

1. *Cells states*: in addition to discrete case, it is possible that cells have a *continuous* state [35], where instead of a finite number of states, cells may have any value in a specific range;
2. *Cells states update*: In many epidemiological models, the states are synchronously updated for the next time step, usually at the end of current time step [36,37]. When cells are updated in different moments throughout the time step, we have an *asynchronous* update [38];
3. *Transition rules*: which can be *deterministic* (elementary CA) or *probabilistic* [39].

The CA used in this paper is Continuous, Asynchronous and Probabilistic, so we have a Continuous Asynchronous Probabilistic Cellular Automata (CAPCA).

It is noteworthy that CA emerged as a flexible tool for numerical simulation of physics and mechanical processes [33,40]. Thereby, it has been also used in population dynamics where local rules and spatial dispersion must be considered in the problem. Usually, urban dynamics are modeled by using CA for dealing with information spreading [41], urban growth [42,43], and urban development [44,45]. On the other hand, the way how individuals interact can also be modeled by using complex random networks [46], which adds a more realistic approach for many local and some long range interactions between individuals of a population. All these models have somehow characteristics of agent-based models [47].

In this paper, CA is a simple and effective tool for enabling spatial dispersion, characteristics of complex networks due to neighbor configuration [37], and could also be described considering agent-based methods. However, the numerical process requires many scenarios to evaluate the model, demanding some processing time.

3. Supervised multi-objective principal component analysis for variable selection

Principal Components Analysis is an efficient and widely used tool for dimensionality reduction of interrelated variables data. It has been repeatedly used as a basis for variable selection approaches in large databases [48–52]. Originally, the objective is to select a subset of original variables, though not considering the relationship between input and output variables.

Recently, some papers have proposed supervised PCA-based strategies when training labels are available [50–53]. The authors in [53], for example, have combined two of the approaches described in [54], with a previous PCA based method proposed in [48], to create a supervised approach that performs an initial selection of variables based on the values of the regression coefficients, without considering the principal components. Moreover, the calculation of the regression coefficients is performed considering a single response variable [53]. Here we propose a supervised method based on PCA to select a set of the most relevant input variables regarding a group of output variables in a multi-objective manner.

The proposed approach applies the PCA in a matrix S created from the relation between inputs and outputs, a *Supervised PCA* (SPCA). The values of S are calculated as standardized regression coefficients $s_{i,j}$ for each input and output variables X_j and Y_i, respectively. The standardized regression coefficients $s_{i,j}$ are computed by the following expression, according to [53]:

$$s_{i,j} = \frac{X_j^T Y_i}{\sqrt{X_j^T X_j}}$$

where X_j is the jth input variable and Y_i the ith output variable.

The PCA is executed over the dataset S to create a matrix S' as a representation of S in the principal component space. So, a contribution index of each input variable X_j as a sum of the scalar products between the jth column of S, S_j, and each column of S' weighted by the corresponding eigenvalues, is calculated considering the first few (main) principal components [55].

4. The model

The population is based on a continuous asynchronous probabilistic cellular automata in the form of a lattice $n \times n$, with $N = n^2$ cells. In order to avoid border effects, the cells from the right border are connected to the cells on the left border, and the same for top and bottom cells, creating a toroidal surface. An individual i has an quantity of information $I_i(t)$ at instant t, which is continuous in interval $0 \leq I_i(t) \leq 1$. When $I_i(t) = 0$, the individual knows nothing about the rumor, and when $I_i(t) = 1$, the individual knows everything and spreads to his neighbors when contacted. Therefore, the individual's state is continuous. The simulations runs for t_s time steps.

Individuals interact with C neighbors inside a Moore radius r. If $C = 4$ and $r = 1$, there will be four interactions in the only possible layer, which is the cells inside radius $r = 1$. For $r > 1$, to interact with layer l, the probability is $q_{l,r} = 2(r + 1 - l)/r(r + 1)$, that is, for $r = 3$, the probability for interacting with $l = 1$ is $q_{1,3} = 1/2$, with

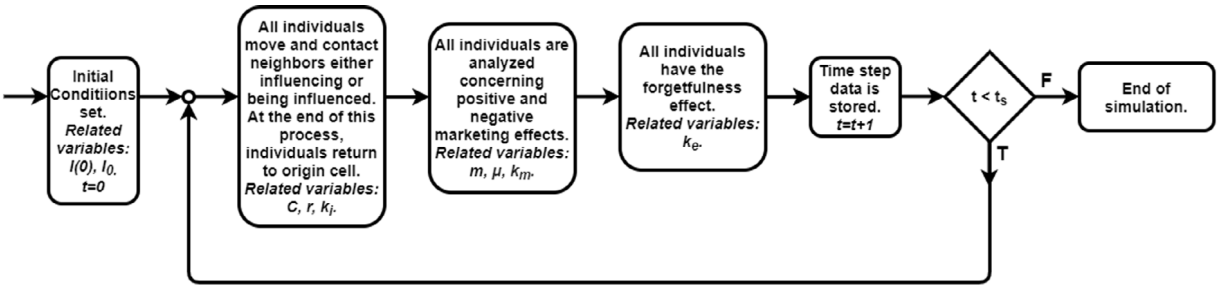
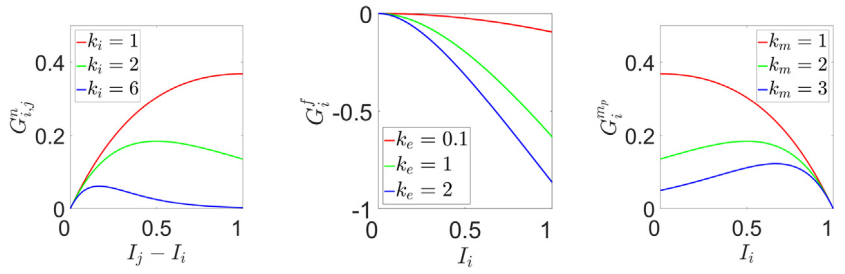


Fig. 1. Diagram of the simulation. After the initial conditions set, all individuals interact with neighbors, and gain/lose information based on interactions, marketing campaigns and forgetfulness. The simulation runs for t_s time steps.



(a) $G^n_{i,j}$ in function of $I_j - I_i$ **(b)** G^f_i in function of I_i **(c)** $G^{m_p}_i$ in function of I_i

Fig. 2. Information gain of an individual I_i for (a) neighbor influence $G^n_{i,j}$, (b) forgetfulness G^f_i and (c) positive marketing $G^{m_p}_i$ for parameters $C = 4$, $r = 4$, (a) $k_i = 1$, $k_i = 2$, $k_i = 6$, (b) $k_m = 1$, $k_m = 2$, $k_m = 3$ and (c) $k_e = 0.1$, $k_e = 1$, $k_e = 2$.

$l = 2$ is $q_{2,3} = 1/3$, and with $l = 3$ is $q_{3,3} = 1/6$. After determined the layer, an randomly individual of that layer is chosen to interact. Such flexibility of the neighborhood allows a higher variability of the population topological structure, from the typical configuration $C = 8$, $r = 1$ to high number of connections in long distances $C = 24$, $r = 20$, for instance [37].

During a time step, the following events occur to individuals (in this order):

- contact C individuals and change information with them;
- receive the effects of positive marketing;
- receive the effects of negative marketing;
- forget part of the information.

The individual's information is updated as soon as the event happens, that is, the information level I_i of an individual may change at each of the C contacts, and at the other events occurrences. Therefore, the state of the individual is not updated synchronously at the end of each time step, and this is the reason for classifying the system as asynchronous.

These events have the following properties:

- influence of a neighbor j: when two individuals have different information levels, the one with more information "teaches" a fraction of information to the other. Consider that $I_j > I_i$ (without loss of generality) and the information gain for i is given by $G^n_{i,j}(t) = I_j(t).(I_j(t) - I_i(t)).e^{-k_i.(I_j(t)-I_i(t))}$, and the information level for j is not modified. The parameter k_i adjusts how flexible is the teaching;
- positive marketing: a percentage m of the population are submitted to a marketing campaign. The information acquired through this process is a function of the current level. Therefore, the gain is $G^{m_p}_i(t) = (1 - I_i(t)).e^{-k_m/(1-I_i(t))}$, and k_m can be adjusted according to a campaign configuration;
- negative marketing: whether an individual is submitted to a marketing campaign for two time steps in a row, a part of the information is lost according to $G^{m_n}_i(I_i, t-1) = \mu.I_i.G^{m_p}_i$, where μ is the intensity of the rejection, with $0 \leq \mu \leq 1$;
- forgetfulness: all individuals forget a part of their information. This lost is given by $G^f_i(t) = -I_i(t).(1 - e^{-k_e.I_i(t)})$, where k_e controls the amount of information lost in function of the current level of the individual.

A diagram summarizing the simulation is shown in Fig. 1.

Note that k_i, k_m and k_e can be either adjusted or tuned depending on the type of the information being modeled in the population. The Fig. 2 shows how these variables have influence over the information variation for a single individual for 2(a) one neighbor influence $G^n_i(I_j - I_i)$, 2(b) the forgetfulness and, 2(c) positive marketing effect. For a hoax, the neighbors influence is high ($k_i = 1$, for instance), the information is easily forgotten ($k_i = 2$, for instance), and there is no marketing campaign ($k_i \gg 1$, for instance). The initial condition of the simulation is based on two variables: I_0 is the maximum information for an individual at $t = 0$, and $\overline{I(0)}$ is the average information of the population at $t = 0$. In the simulations, individuals start with a random value in the interval $0 < I_i(0) < I_0$.

These equations were designed to formalize any information which could spread in the population. Moreover, they replicate some dynamics from other models for gaining information through individual contact or press [2,3,18], with marketing campaigns [23,24], and losing information due to forgetfulness [3,7,18] and marketing rejection [25].

Also, note that such separated processes enable the model to be set according to the type of information being used. For instance, a pure rumor based on a word-of-mouth practice could have $k_m = 0$ with $k_e \gtrsim 1$, indicating information gain based only on neighbor influence and a high rate of forgetfulness. A viral marketing campaign would have $k_i = 0$, and $k_m \gtrsim 1$, indicating high influence of advertisement.

The Table 1 contains a summary of the variables.

5. Results

The population configuration used in the simulations are $n = 100$, $N = 10\,000$, $C = 4$ and $r = 4$. The initial condition is based on I_0 and $\overline{I(0)}$, which are described for each case. Moreover, only individuals from a square 10×10 centered in the middle of the lattice have

Table 1
Notation used in the paper.

Table of notation		
Notation	Definition	Description
C	$\in \mathbb{N}^+$	Individual's interactions with its neighbors.
r	$\in \mathbb{N}^+$	Maximum radius for interactions.
$G^n_{i,j}$	$\in [0,1]$	Information gain through neighbor influence.
$G^{m_p}_i$	$\in [0,1]$	Information gain through positive marketing.
G^f_i	$\in [-1,0]$	Information variation due to forgetfulness.
$G^{m_n}_i$	$\in [0,1]$	Information variation due to negative marketing process.
k_i	$\in [0,\infty]$	Parameter related to $G^n_{i,j}$.
k_m	$\in [0,\infty]$	Parameter related to $G^{m_p}_i$.
k_e	$\in [0,\infty]$	Parameter related to G^f_i.
m	$\in [0,1]$	Percentage of individuals which have positive marketing influence.
μ	$\in [0,1]$	Rejection to marketing campaigns, related to negative marketing process.
$\overline{I(0)}$	$\in [0,1]$	Average information in population as initial condition.
$\overline{I_0}$	$\in [0,1]$	Maximum information for an individual as initial condition.
t_s	$\in \mathbb{N}^+$	Simulation time steps.

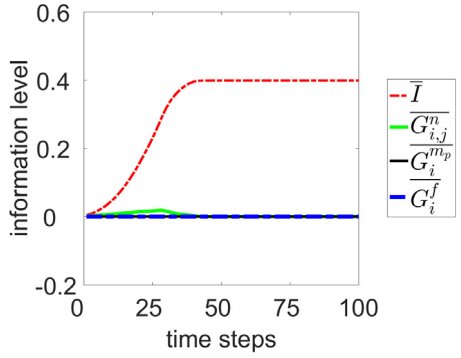


Fig. 3. Time evolution of information level ($\overline{I}$), neighborhood influence (G^n_i), positive marketing ($G^{m_p}_i$), forgetfulness (G^f_i) and variation due to death process (G^d_i) by using the parameters $C = 4$, $r = 4$, $k_i = 1$, $k_m = 0$, $k_e = 0$, $m = 0$, $\mu = 0$, $\overline{I(0)} = 0.002$ and $I_0 = 0.4$; moreover, all individuals i with $I_i(0) > 0$ are concentrated in a square 10×10 in the center of the lattice.

$I_i(0) > 0$, with the rest of population starting with $I_i(0) = 0$. This is based on the fact that rumors usually start to spread from a small group of individuals. Higher values of C and r would make the information to spread faster throughout the population. These values have been chosen to allow a randomness neighborhood since each individual will connect to 4 of 80 neighbors, and to control the required time to achieve the permanent regime. Thus, it will be possible to compare information gaining and losing processes.

Each simulation runs for $t_s = 100$ time steps. For all the simulations and tests with the model, the system reached the permanent regime before time step $t = 70$, then all the output variables are considered as an average value for the last 20 time steps, where applied. Note that in the permanent regime, the total information gained is equal to the total of information lost per time step.

Initially, using $k_i = 1$, $k_m = 0$, $k_e = 0$, $m = 0$, $\mu = 0$, $\overline{(I(0))} = 0.002$ and $I_0 = 0.4$, we have only the spread of information by neighbor influence without forgetfulness and marketing. Fig. 3 have the time evolution and Fig. 4 the spatial spreading of the information. The average gain by neighbor influence is small, but since it happens for the whole population of $N = 10\,000$ individuals, all of them have the information level close to $I_0 = 0.4$ in the permanent regime.

By adding the forgetfulness with $k_e = 0.25$, the information spreads throughout the lattice, but the level of information in the population remains constant, what is seen in Figs. 5 and 6. That is, the information level is constant, but it is diluting in the individuals and spreading in the space. Finally, Figs. 7 and 8 have the results for the simulation with a marketing campaign configured with $m = 0.1$, $k_m = 1$ and $\mu = 0.5$ (we also changed the forgetfulness parameter to $k_e = 0.5$). The first thing to

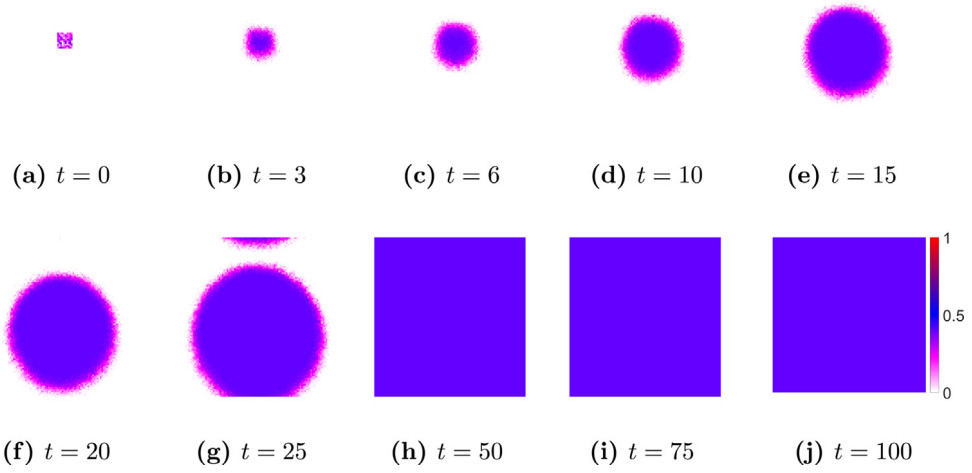


Fig. 4. Spatial propagation of the case represented in Fig. 3 for time steps $t = 0$, $t = 3$, $t = 6$, $t = 10$ $t = 15$, $t = 20$, $t = 25$, $t = 50$, $t = 75$ and $t = 100$. The color represents the information level of each cell with the values in the colorbar right to the last figure.

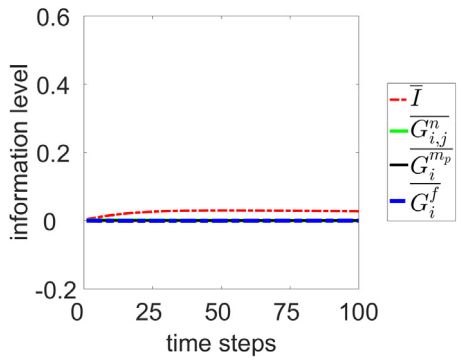


Fig. 5. Time evolution of information level ($\overline{I}$), neighborhood influence (G^n_i), positive marketing ($G^{m_p}_i$), forgetfulness (G^f_i) and variation due to death process (G^d_i) by using the parameters $C = 4$, $r = 4$, $k_i = 1$, $k_m = 0$, $k_e = 0.25$, $m = 0$, $\mu = 0$, $\overline{I(0)} = 0.002$ and $I_0 = 0.4$.

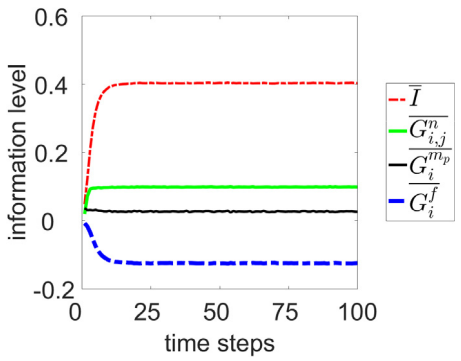


Fig. 7. Time evolution of information level ($\overline{I}$), neighborhood influence (G^n_i), positive marketing ($G^{m_p}_i$), forgetfulness (G^f_i) and variation due to death process (G^d_i) by using the parameters $C = 4$, $r = 4$, $k_i = 1$, $k_m = 1$, $k_e = 0.5$, $m = 0.1$, $\mu = 0.5$, $\overline{I(0)} = 0.002$ and $I_0 = 0.4$.

note is that there is not the wave of information spreading in the lattice because all the individuals have the same chance to receive information through the marketing campaign — also, a higher level of information in the population by marketing results in higher level of forgetfulness.

Concerning the marketing, variables k_m and m are related to the decision processes to determine a marketing campaign. How many individuals will have contact with the campaign is controlled by m and the kind of information to be given to individuals is controlled by k_m. In order to understand their impact in the population dynamic, Figs. 9 and 10 have the average marketing gain ($\overline{G^{m_p}} - \overline{G^{m_n}}$) and the average information level ($\overline{I}$), respectively. These variables are plotted in function of m and k_m for all the combinations of the parameters $k_i = 2, 6, 10$ and $k_e = 0.1, 1, 2$ using $\mu = 0.5$, $\overline{I(0)} = 0.001$ and $I_0 = 0.2$. There were one simulation for each combination of $m = 0, 0.02, \ldots, 1$ and $k_m = 0, 0.02, \ldots, 3$, resulting in 7701 cases, where the average values of the last 20 time steps for $\overline{G^{m_p}} - \overline{G^{m_n}}$ and $\overline{I}$ is considered to generate the figure.

By analyzing the figures, when the forgetfulness is low ($k_e = 0.1$), the information level is high, and the marketing gain is low because the population already have this high level of information. For higher forgetfulness in the population ($k_e = 0.1$), the marketing gain is maximum when $k_m = 0$ and $m \approx 0.5$. However, the higher average information level occurs when $k_m \approx 1$ and $m \approx 0.45$.

The last simulation to be shown is the effect of the marketing rejection in Fig. 11. Therefore, $\overline{I}$ (11a, b) and $\overline{G^{m_p}} - \overline{G^{m_n}}$ (11c, d) are plotted in function of μ and m with $k_m = 1$ (11a, c) and in function of μ and k_m with $m = 0.3$ (11b, d) using $\overline{I(0)} = 0.002$, $I_0 = 0.4$, and $k_i = 3$. Again, independent of μ, the marketing gain and the information level have a higher values when $m = 0.5$ with low k_m.

Before discussing these results, the SPCA method is applied to simulation data and presented in the next section.

6. Statistical analysis with SPCA

Since the model has five input variables related to information variation at individual level, it is hard to precisely see which one is more relevant to the model output only by using numerical simulations. Therefore, 10,000 simulations have been done to check what are the most relevant input variables of the model using the SPCA model presented in Section 3. For all simulations, the values of k_i, k_m, k_e, m and μ were randomly chosen for the intervals: $0 \le k_i \le 10$, $0 \le k_m \le 10$, $0 \le k_e \le 10$, $0 \le m \le 1$, $0 \le \mu \le 1$. The output variables used were: $\overline{I}$, $\overline{G^n}$, $\overline{G^{m_p}}$, $\overline{G^{m_n}}$ and $\overline{G^f}$. Note that the average values of these variables are calculated by considering only individuals that had a gain/loss due to the process. For instance, if the total gain for positive marketing in the population is $G^{m_p} = 50$, and 500 individuals received the marketing campaign, then $\overline{G^{m_p}} = 0.1$.

The proposed approach is classified as a supervised method because it selects the most relevant input variables based upon its relationship with the output variables. This relationship is represented by the matrix of standardized regression coefficients S, which is calculated by the relation between the inputs k_i, k_m, k_e, m and μ and the outputs I, G^n,

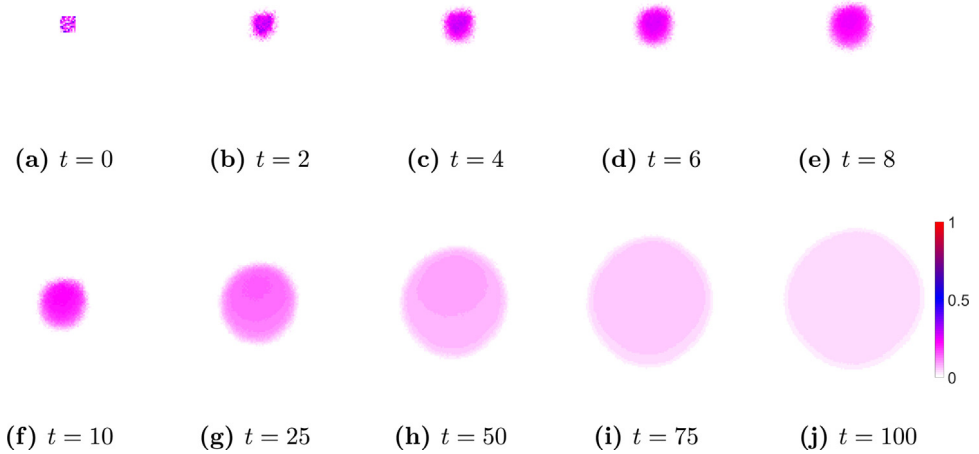


Fig. 6. Spatial propagation of the case represented in Fig. 5 for time steps $t = 0$, $t = 2$, $t = 4$, $t = 6$ $t = 8$, $t = 10$, $t = 25$, $t = 50$, $t = 75$ and $t = 100$. The color represents the information level of each cell with the values in the colorbar right to the last figure.

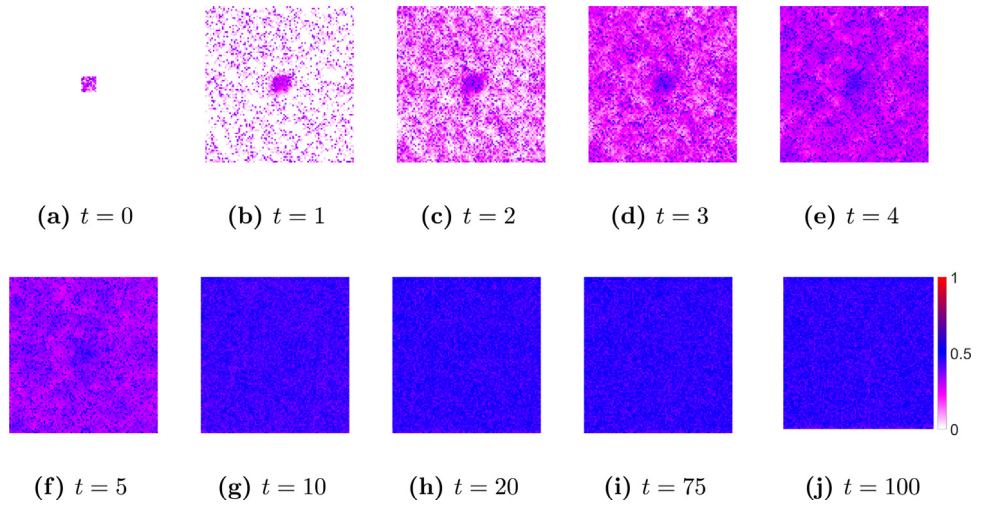


Fig. 8. Spatial propagation of the case represented in Fig. 7 for time steps $t = 0$, $t = 1$, $t = 2$, $t = 3$ $t = 4$, $t = 5$, $t = 10$, $t = 20$, $t = 75$ and $t = 100$. The color represents the information level of each cell with the values in the colorbar right to the last figure.

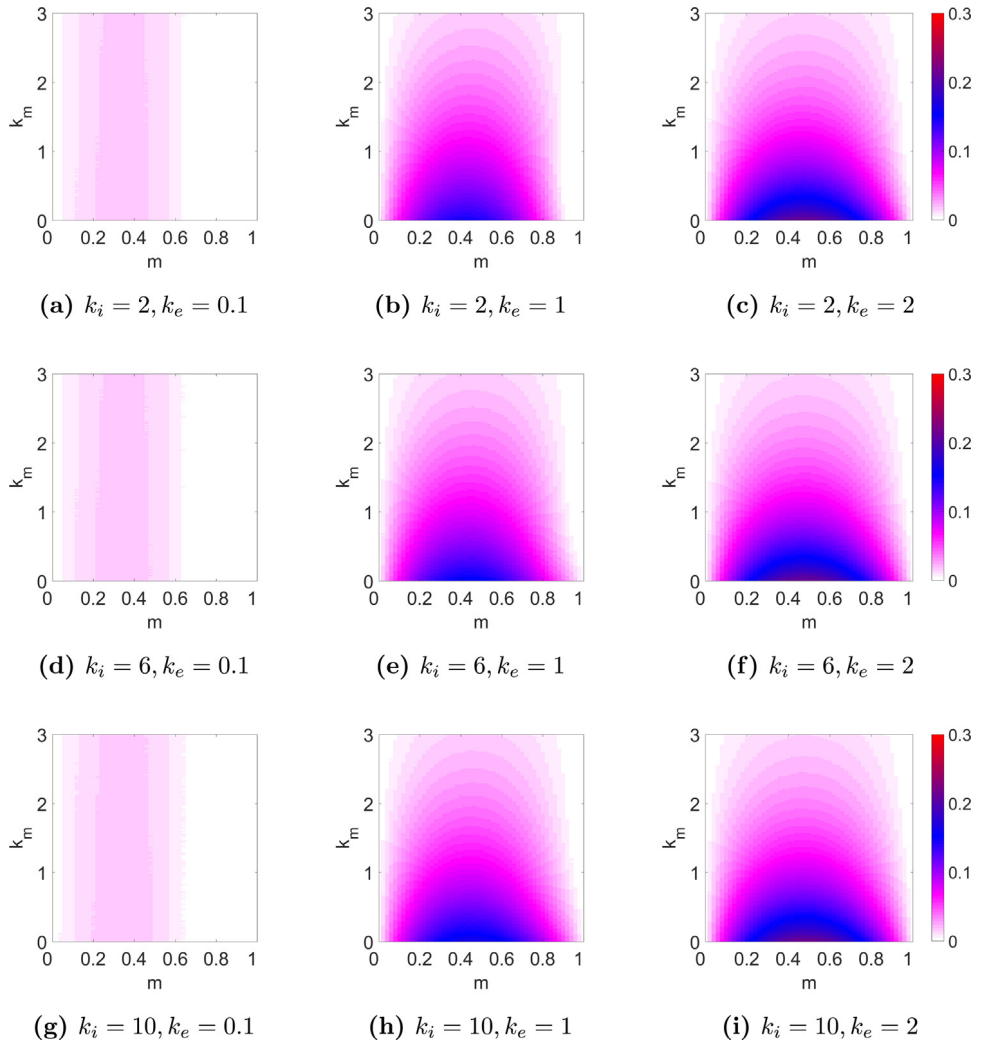


Fig. 9. Marketing gain $(\overline{G^{m_p}} - \overline{G^{m_n}})$ in function of m and k_m for $k_i = 2, 6, 10$ and $k_e = 0.1, 1, 2$ using $\mu = 0.5$, $\overline{I(0)} = 0.001$ and $I_0 = 0.2$. The colorbar represents the color for different values of $\overline{G^{m_p}} - \overline{G^{m_n}}$.

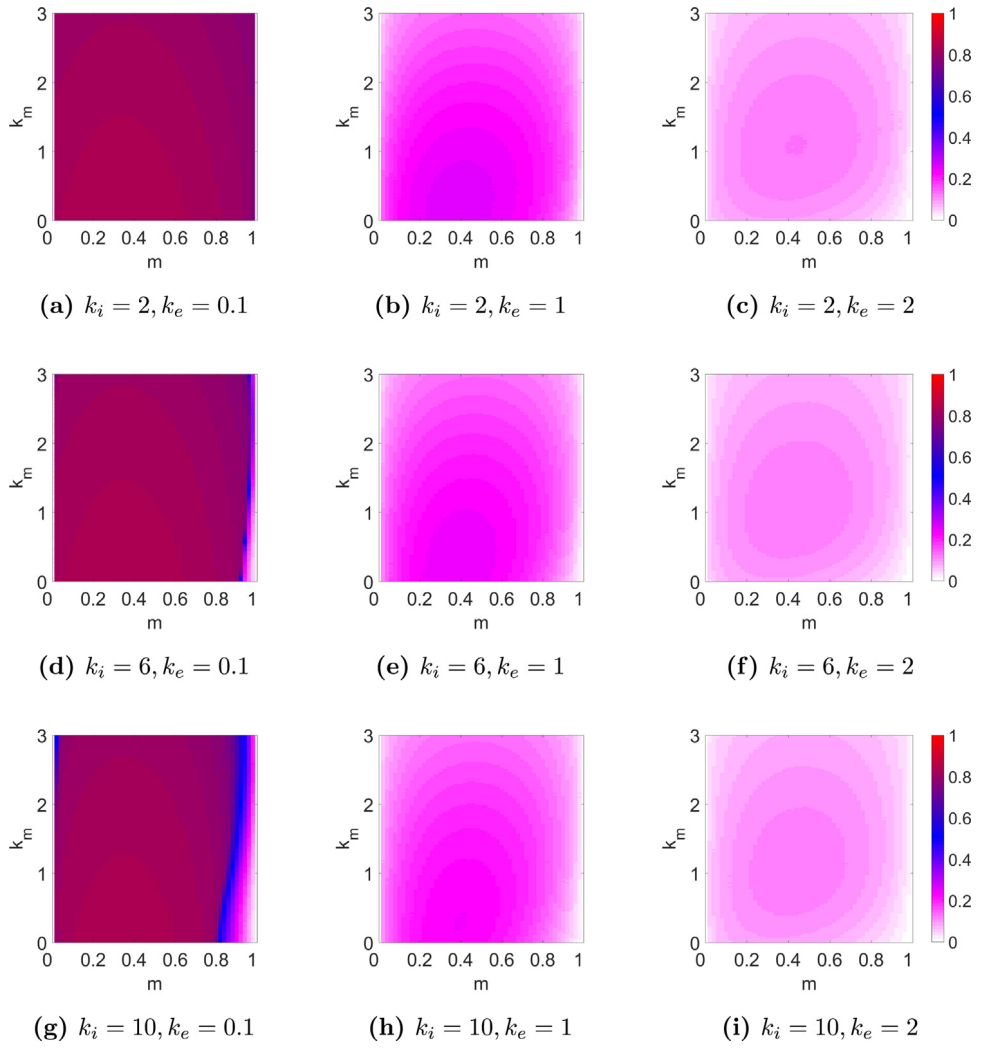


Fig. 10. Information level ($\overline{I}$) in function of m and k_m for $k_i = 2, 6, 10$ and $k_e = 0.1, 1, 2$ using $\mu = 0.5$, $\overline{I(0)} = 0.001$ and $I_0 = 0.2$. The colorbar represents the color for different values of $\overline{I}$.

G^{m_p}, G^{m_n} and G^f, as follows:

$$S = \begin{bmatrix} \frac{k_i^T I}{\sqrt{k_i^T k_i}} & \frac{k_m^T I}{\sqrt{k_m^T k_m}} & \frac{k_e^T I}{\sqrt{k_e^T k_e}} & \frac{m^T I}{\sqrt{m^T m}} & \frac{\mu^T I}{\sqrt{\mu^T \mu}} \\ \frac{k_i^T G^n}{\sqrt{k_i^T k_i}} & \frac{k_m^T G^n}{\sqrt{k_m^T k_m}} & \frac{k_e^T G^n}{\sqrt{k_e^T k_e}} & \frac{m^T G^n}{\sqrt{m^T m}} & \frac{\mu^T G^n}{\sqrt{\mu^T \mu}} \\ \frac{k_i^T G^{m_p}}{\sqrt{k_i^T k_i}} & \frac{k_m^T G^{m_p}}{\sqrt{k_m^T k_m}} & \frac{k_e^T G^{m_p}}{\sqrt{k_e^T k_e}} & \frac{m^T G^{m_p}}{\sqrt{m^T m}} & \frac{\mu^T G^{m_p}}{\sqrt{\mu^T \mu}} \\ \frac{k_i^T G^{m_n}}{\sqrt{k_i^T k_i}} & \frac{k_m^T G^{m_n}}{\sqrt{k_m^T k_m}} & \frac{k_e^T G^{m_n}}{\sqrt{k_e^T k_e}} & \frac{m^T G^{m_n}}{\sqrt{m^T m}} & \frac{\mu^T G^{m_n}}{\sqrt{\mu^T \mu}} \\ \frac{k_i^T G^f}{\sqrt{k_i^T k_i}} & \frac{k_m^T G^f}{\sqrt{k_m^T k_m}} & \frac{k_e^T G^f}{\sqrt{k_e^T k_e}} & \frac{k_i^T G^f}{\sqrt{m^T m}} & \frac{\mu^T G^f}{\sqrt{\mu^T \mu}} \end{bmatrix}$$

$$= \begin{bmatrix} 1.348 & 0.826 & 0.546 & 1.367 & 1.373 \\ 0.027 & 0.016 & 0.014 & 0.034 & 0.036 \\ 0.147 & 0.030 & 0.148 & 0.144 & 0.144 \\ -0.008 & -0.002 & -0.003 & -0.011 & -0.010 \\ 0.165 & 0.044 & 0.159 & 0.167 & 1.169 \end{bmatrix}$$

It is important to note that a column of S corresponds to the relation between a single input variable and all output variables. The PCA is then applied to S and the main components are selected to compute their representations in the principal component space, S'. In this case, only one principal component has been selected.

$$S' = \begin{bmatrix} 2.559 \\ 0.060 \\ 0.271 \\ -0.017 \\ 0.313 \end{bmatrix}$$

And the contribution index of each variable is:

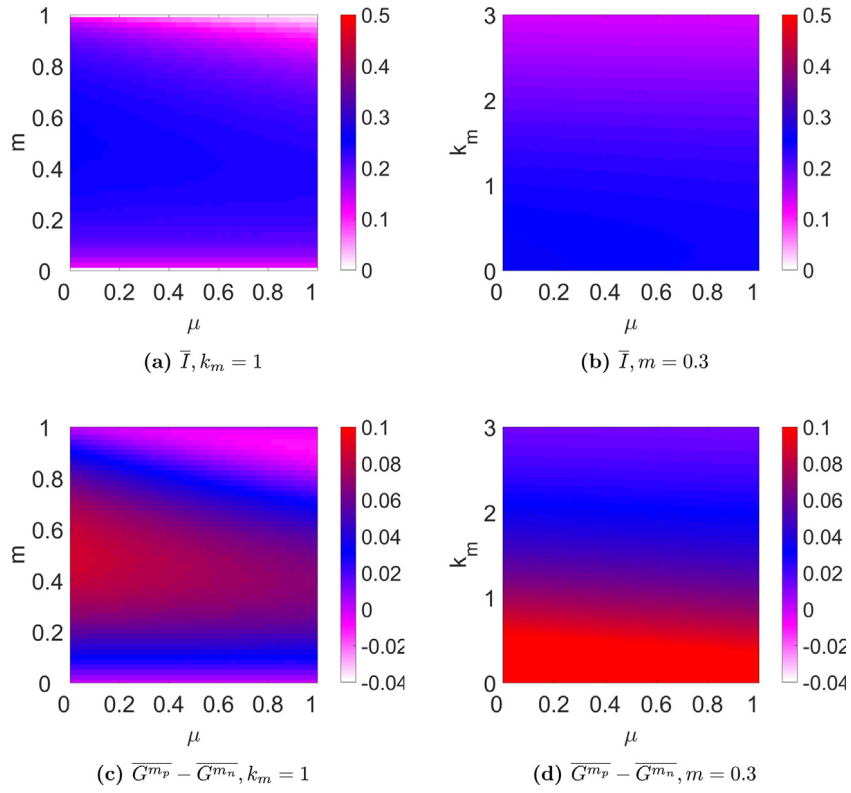
$$C_I(k_i, k_m, k_e, m, \mu) = (0.982, 0.592, 0.412, 0.995, 1.000)$$

as illustrated in Fig. 12.

According to SPCA, μ, m and k_i are the most important variables for the variance of the model output. It means that these three variables exerts the strongest influence on the responses, with k_e being the variable which has a weak influence over the output.

7. Discussion

In this paper, we presented a rumor propagation model in a population-based on continuous asynchronous probabilistic cellular automata, where marketing campaigns can be set to boost the rumor spread. However, as results indicated, the parameters for marketing need to be carefully set according to information and population characteristics. According to the model, for a range of the parameters,



(a) $\overline{I}, k_m = 1$ (b) $\overline{I}, m = 0.3$

(c) $\overline{G^{m_p}} - \overline{G^{m_n}}, k_m = 1$ (d) $\overline{G^{m_p}} - \overline{G^{m_n}}, m = 0.3$

Fig. 11. Information level ($\overline{I}$) [a, b] and marketing gain ($\overline{G^{m_p}} - \overline{G^{m_n}}$) [c, d] in function of rj and m with $k_m = 1$ [a, c] and in function of μ and k_m with $m = 0.3$ [b, d] using $\overline{I(0)} = 0.002$ and $I_0 = 0.4$. The colorbar represents the color for different values of $\overline{I}$ in a, b, and $\overline{G^{m_p}} - \overline{G^{m_n}}$ in c, d.

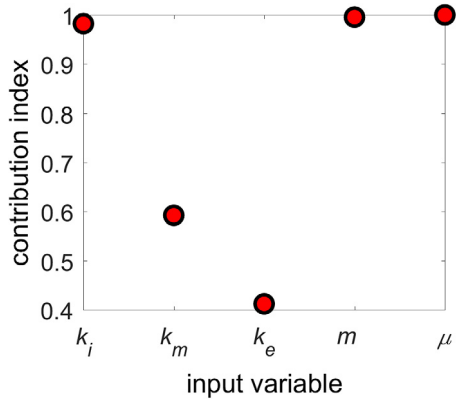


Fig. 12. Contribution index of each input variable regarding the effect on the output variables.

choose an intermediary value for m, and k_m is a better strategy for maximizing the information level on the population, independent of the marketing rejection.

However, determine how a population deal with a rumor and set values to k_i, k_e, and μ is hard. Therefore, by using the proposed method SPCA, we could run 10,000 simulations to have a wide range of parameters variation. In this case, the most critical variables that impact on the output ones are k_i, m and μ, that is, before starting a marketing campaign, the number of individuals who will receive the campaign is a significant factor to maximize the information level in the population, and word-of-mouth cannot be ignored. Also, the weight of the marketing rejection must be taken into account, since it also mostly impacts the information level. The first is a decision and can be easily set for simulations, but for the second it needs some research to see how individuals react to that kind of campaign. A vaccination campaign probably has a low rejection, but a bank advertisement can have a high rejection if it is too exposed to the media.

The results corroborates the significant importance for the word-of-mouth during a marketing campaign like reported in [24]. Even using a heterogeneous population, the authors also showed the importance of the word-of-mouth during marketing campaigns because individuals tend to trust more on relatives and friends. Also, companies formulate marketing campaigns simulating processes of word-of-mouth in an attempt to mimic this trust [28], usually using celebrities who speak unpretentiously about the product on their social networks. It is worth mentioning that online consumer review has a similar impact of word-of-mouth in buying processes [56].

Since the model is based on marketing purposes for information spreading, the rejection plays a significant role and must be considered as a possible side effect. Marketing clutter and intrusiveness have been studied to balance the needs of the advertiser and the audience. For instance, on [31], the authors showed the negative impact of clutter on entertainment magazines, discussing where the marketing starts to have negative effects on the magazines' revenues. Such threshold has also been studied when considering website advertisements [30,32,57], however, negative effects do not happen on conspiracy rumors on the internet, that only get stronger and more propagated in function of the time [58]. This paper proposed a model to evaluate a marketing campaign result, and by setting variables k_i, k_m, and k_e for information type, and C, and r for interaction characteristics of the population (or using a complex network model), it could be possible to balance marketing variables (k_m and m) to maximize the return.

Another contribution of this work was the formulation of the model based on cellular automata due to flexibility to consider spatial information spreading, and a neighborhood configuration [37] that allows setting the population with complex network characteristics. Cellular automata have been used for modeling the spatial distribution of individuals altogether with a variety of problems, like disease propagation [37], information spreading [41], consumer market behavior [6],

and urban evolution [44]. Information spreading on urban spaces had some studies in order to analyze transit advertising [41] and diffusion of false information during public crisis [43], and cellular automata provided the flexibility required for these type of problems. The model presented in this paper could be used on urban space with the following modifications: each cell could contain more than one individual to allow different population densities on the map, some central individual could interact more than others to simulate different influential characters, and use a map to create the cellular automata spatial distribution.

However, before a increasing of the model complexity, the next steps of this work can go in the following directions with the current configuration:

- use a heuristic algorithm to tune k_i, k_m and k_e based on real data of a rumor;
- test the model on complex networks and;
- propose a variation of the model to input a control variable to prevent the rumor to spread.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

E. Silva: Software, Investigation, Visualization. **F.H. Pereira:** Validation, Formal analysis, Data curation, Writing - review & editing. **P.H.T. Schimit:** Conceptualization, Methodology, Writing - original draft.

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